

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1. We seek to approximate  $\int_0^2 x^2 dx$  using the rectangular rule, with  $w = 0.5$ . For convenience, the data you will need to perform this computation are presented in the table:

1 / 1 point

| x    | $g(x)$ |
|------|--------|
| 0.25 | 0.0625 |
| 0.75 | 0.5625 |
| 1.25 | 1.5625 |
| 1.75 | 3.0625 |

What is the approximate integral computed by the rectangular rule? Enter your answer with three digits to the right of the decimal point.

2.625

✔

Correct

Yes, this is correct. Note that the true integral value is 2.6666...

2. We seek to approximate  $\int_0^2 x^2 dx$  using the trapezoid rule, with  $w = 0.5$ . For convenience, the data you will need to perform this computation are presented in the table:

1 / 1 point

| x   | $g(x)$ |
|-----|--------|
| 0   | 0      |
| 0.5 | 0.25   |
| 1.0 | 1.0    |
| 1.5 | 2.25   |
| 2.0 | 4.0    |

What is the approximate integral computed by the trapezoid rule? Enter your answer with two digits to the right of the decimal point.

2.75

✔

Correct

Yes, this is correct.Note that the true integral value is 2.6666...

3. Which approximation to an integral do we expect to be the more accurate when the region width  $w$  is small?

1 / 1 point

- ☐

The rectangular rule.
- ☒

The trapezoid rule.

✔

Correct

Yes, this is correct.